THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

Department of **Accounting**

Five Days at Fremont — Part A

AC101 integrated Tesla case · worked in Lecture 1

Before you start: what is real here, and what is not

This case is set inside a real company; it consists of both fact and fiction.

- **Tesla, Inc. is real**, and some of the financial data below comes from actual SEC filings and Tesla investor-relations reports; the period is stated on each exhibit. **All product costing scenarios, internal cost data, standards and strategic decisions in this case are entirely hypothetical.** They do not represent actual Tesla plans, costs or decisions, and any resemblance beyond the public filings is coincidental. The CFO, the colleagues and all internal dialogue are fictional; no real Tesla executive is depicted.
- **Every exhibit is labelled** — **REAL** (SEC 10-K FY2024, Compustat 2023–24, or Tesla IR, with the period stated), **SIMULATED** (with a one-line note on how the figures were constructed — industry benchmarks, teardown reports, the BloombergNEF battery index, or standards built on the real Q4 2024 base), or **ESTIMATE** (a statistic we computed from the real data, such as Part B's regression — not a figure Tesla discloses).
- **One stylised liberty, flagged where it is used (Part C):** all five models are placed at the Fremont factory for a clean single-factory example. In reality the Cybertruck is built at Giga Texas. Part C's margins are stylised levels — roughly 2.4 times Tesla's real 17.9% FY2024 gross margin — so read the *distribution* across models, not the level.
- **Battery figures in Part D are cell-level variable costs.** Do not compare them with pack-level prices (~$115–140/kWh, BloombergNEF 2024), which add pack assembly, housing and battery management on top of the cell — a different scope.
- **Proxies are labelled.** "Revenue per car delivered" spreads energy, services and regulatory-credit revenue over cars delivered — it is *not* an average selling price and sits above the ~$43k Model Y sticker. Part B's regression is estimated on Compustat's COGS series; Compustat reports COGS with depreciation, amortisation and impairment shown separately ($5,368m in FY2024), so the intercept understates Tesla's total fixed production cost. Part C's discontinuation reload is a labelled worst case.

How to use this case. One part per day: each lecture ends with a case block — read that day's part, work its questions in your groups, and we discuss the solutions together. Every question is discussed in the lecture. A ▶ marks a more advanced question. A calculator is all you need; where figures are in $m, the question says so.

One note on the exam. Variance analysis (Part E) is taught in Part 1 but **examined at the final exam only** — it does not appear on the mid-session paper. Everything else here is mid-session material. Any sales-volume variance in this case is **measured on a contribution basis**, and says so.

Monday, 7:40 a.m.

The Fremont factory is louder than the recruitment video suggested. You are a newly hired analyst on Tesla's corporate finance team, on site for your first week, and you have been inside the plant for forty minutes — long enough to watch a robot lift a painted body shell onto a battery pack in one unhurried motion — when your phone buzzes.

It is the CFO. Call her Reyes; like every executive in this case, she is invented.

> *"Welcome aboard. Simple question for your first week: are we making money on the Model Y? Briefing Friday, 4 p.m. Show your working."*

It reads like a question with a single number for an answer. By Friday afternoon you will know that it is five different questions wearing one sentence — and that each of them needs a different "cost".

Tesla in 2024: the company behind the question

Tesla designs, builds and sells electric vehicles and energy products, selling directly to customers rather than through franchised dealers. The Fremont, California plant — a former GM–Toyota joint-venture factory that Tesla bought in 2010 — builds the Model S, Model 3, Model X and Model Y; the Cybertruck is built at Giga Texas.

The financial year you have walked into was bruising. In fiscal 2024 Tesla reported total revenues of **$97.69bn** and delivered **1,789,226** vehicles (producing 1,773,443) — but the first quarter of 2024 brought the company's first year-on-year delivery decline since the pandemic quarter of Q2 2020, and a price war that began with a single $13,000 cut to the Model Y in January 2023 was still grinding on. Growth was no longer doing the explaining. Cost had to.

That is the world your Monday message landed in.

Part A — Monday: cost of *what*?

Your first instinct is the obvious one: look it up. Tesla files audited accounts with the SEC; surely the cost of a Model Y is in there somewhere.

It is not — and discovering *why* it is not is the real work of Day 1.

The FY2024 income statement (Exhibit A-1) is strikingly compact. Revenue of $97,690m; cost of revenues of $80,240m; gross profit of $17,450m; then $5,150m of selling, general and administrative expense, $4,540m of research and development, $684m of restructuring, and an operating income of $7,076m. Six numbers below the revenue line, for a company that builds cars, batteries, software and Superchargers on three continents. The accounts are built for *outsiders* — investors, lenders, regulators — who need standardised, comparable, backward-looking totals. They are financial accounting. Reyes's question is an *insider's* question — forward-looking, decision-shaped, and about one product — and answering it is management accounting, which the published accounts were never designed to do.

So you start where a management accountant starts: **cost of what?** A cost only means something once you fix the **cost object** — a product (one Model Y), a plant (Fremont), a customer (a rental fleet), a project (Cybertruck development). The same salary is "direct" to the factory but "indirect" to a single car. Object first, cost second.

Then you sort costs two ways, and each way exists because it answers a different manager's question:

- **By behaviour — "how does profit move with volume?" Variable costs** change in total with output (battery cells: more cars, more cells). **Fixed costs** stay flat in total across a sensible range of volumes (factory insurance costs the same at 400,000 or 500,000 cars).
- **By traceability — "what does this product cost?" Direct costs** can be traced economically to the cost object (the battery pack in *this* Model Y). **Indirect costs — overhead —** are shared and must be *allocated* by some rule (the plant manager's salary covers every model). How to allocate is Wednesday's problem, and it is a big one.

A colleague on the costing team, hearing it is your first day, sends you a welcome present: a list of ten Fremont costs "to see whether the new analyst can sort the post" (Exhibit A-2). Every cost sits on *both* dimensions at once — paint is variable *and* direct; the property tax is fixed *and* indirect — and the 2×2 the ten items build is the foundation of the whole week.

One more distinction matters before you can read Tesla's statements properly: **cost versus expense**. A cost is a resource given up to get something; it becomes an expense only when it is *used up* to earn revenue.

Some costs expense immediately (today's advertising). Some expense gradually (a machine, through depreciation). And a car's build cost does something stranger: it waits on the balance sheet as **inventory** — an asset — and hits profit as **cost of goods sold** only when the car is sold. A Model Y built in December and sold in January carries its cost across the year-end as an asset. Nothing is lost; it is just not in *this* year's profit. Hold onto that lane of the cost journey — it comes back with teeth on Wednesday.

Late in the afternoon, a memo lands from another analyst (Exhibit A-3, illustrative round numbers — explicitly not Tesla's actuals). At 300,000 cars, the accountants report a full unit cost of $30,000 per car: $20,000 variable plus $10,000 of fixed cost per car ($3.0bn spread over 300,000). The memo scales that $30,000 to a 500,000-car plan. It looks diligent. It is wrong by two billion dollars — because "fixed cost per car" is an artificial number that slides with whatever volume you divide by ($10,000 at 300,000 cars; $6,000 at 500,000). Total fixed cost does not move; only the arithmetic does. For decisions, think in totals.

By Monday evening you can give Reyes the honest half-answer. **Tesla as a whole made money in 2024: $17.45bn gross, $7.08bn operating** — straight off Exhibit A-1. But "the cost of a Model Y" is not one number. The full average cost (with allocated factory and R&D), the incremental cost of one more car, and the avoidable cost of not making it are *different numbers answering different decisions* — and the published accounts reveal none of them. Recovering them is the rest of your week.

Assignment questions — Part A

1. From Exhibit A-1, express cost of revenues and gross profit as a percentage of revenue. If you were the CFO, which of the three big cost lines — cost of revenues, SG&A, R&D — would you attack first, and why?
2. Cost object: **one Model Y**. Classify each of Exhibit A-2's ten costs on both dimensions — fixed/variable and direct/indirect — with a one-line reason for each. Check yourself against the classification table from Session 1.
3. A Model Y is built in December 2024 but sold in January 2025. At the 31 December year-end, is its build cost a cost or an expense — and where does it sit? When does it hit profit, and under what name?
4. Using Exhibit A-3: a colleague budgets for 500,000 cars by scaling the $30,000 full unit cost. Compute their budget and the correct one, and explain in two sentences where the phantom $2.0bn comes from.
5. ▶ The 10-K alone cannot say whether the Model Y is profitable. Name three different "costs of a Model Y" and match each to a decision it serves, one line each. (This is the map of Parts C and D.)
6. Exhibit A-4 gives one quarter's cost ledger for the Model Y line — and it contains more than you need. **(a)** Cost object: **one Model Y**. Using only the rows that belong, compute the variable cost of building one Model Y (its variable manufacturing cost). List every row you leave out with a one-line reason for rejecting it, and state one assumption your figure relies on. **(b)** ▶ One of the rows you rejected in (a) is a genuine cost of building Model Ys, yet it stayed out of your figure. Name that row, give the one word that justifies leaving it out here, and name the different "cost of a Model Y" it would belong to. No calculation needed.

Exhibits — Part A

Exhibit A-1 — Tesla FY2024 income statement *REAL — SEC 10-K, fiscal year 2024 (SEC EDGAR filing index).*

Line item	$m
Total revenues	97,690
Cost of revenues	(80,240)
Gross profit	**17,450**
Selling, general and administrative	(5,150)
Research and development	(4,540)
Restructuring and other	(684)
Operating income	**7,076**

Exhibit A-2 — Ten Fremont costs to classify *SIMULATED — illustrative cost descriptions only (no figures); a hypothetical exercise, not Tesla's cost records.*

1. Paint and coatings
2. Fremont property tax
3. Tyres
4. Welding gas and machine lubricants (used across all models)
5. Depreciation on Model-Y-only welding robots
6. Plant manager's salary
7. Shipping to the customer
8. Quality-inspector salary
9. Aluminium for body panels
10. Factory canteen

Exhibit A-3 — The analyst's shortcut *SIMULATED — illustrative round numbers chosen to make the mechanics obvious; explicitly not Tesla's actuals.*

Item	Figure
Total fixed cost per period	\$3.0bn
Variable cost per car	\$20,000
Fixed cost per car at 300,000 cars	\$10,000
"Full unit cost" at 300,000 cars	\$30,000
Fixed cost per car at 500,000 cars	\$6,000
The memo's budget for 500,000 cars	500,000 × \$30,000

Exhibit A-4 — One quarter on the Model Y line *SIMULATED — a hypothetical quarter's cost ledger for the Model Y line only, built from teardown reports and industry benchmarks; not Tesla's cost records. Figures are quarterly totals in \$m unless a per-car figure is stated; the battery figure is pack-level (about 75 kWh at benchmark pack cost, consistent with the ~\$115–140/kWh pack prices flagged on the first page).*

Cost line	\$m
Battery cells and pack	1,536
Other direct materials (body, drive units, electronics, glass, tyres, interior)	2,944
Assembly and stamping labour (paid by the hour on the line)	560
Factory building depreciation, insurance and plant-management salaries	640
Delivery, showrooms and marketing for cars sold this quarter	480
Battery-research spending charged to the line this quarter	220
Retooling of the Model Y line, completed and paid last quarter	400
Memo: Model Y cars built this quarter	*160,000 cars*